

Marwan Eid

Software Engineer · Backend & Full-Stack

Giza, Egypt marwan.eid1999@gmail.com [linkedin.com/in/marwan-eid](#) [github.com/marwan-eid](#) [marwan-eid.github.io](#)

WORK EXPERIENCE

- Software Engineer · Valeo

Jul 2023 – Present

 - Designed an event-driven, MQTT-based vehicle communication platform on AWS (EKS, Lambda) for multi-party real-time streaming sessions, placing each session on the least-loaded streaming server and routing device joins through an RDS-backed session registry. Led 2 engineers across system design and delivery.
 - Automated CI/CD pipelines with GitHub Actions, Docker and Helm on Kubernetes, cutting deployment time from hours to under 10 minutes with zero-downtime rollouts and automated rollbacks.
 - Built an event-driven, closed-loop AI system that detects operational anomalies in CloudWatch logs, generates candidate fixes, and verifies each fix in sandboxed local and demo environments before routing it to engineers for review.
 - Cut API response latency from 10 s to 1 s by rewriting cold-start-bound Java Lambda functions in Go.
- Junior Software Engineer · Valeo

Jul 2022 – Jul 2023

 - Incrementally migrated a 75k+ LOC legacy C# desktop application for vehicle lifecycle management (VW, Volvo, HKM) to the web, extracting modular ASP.NET MVC APIs behind a new React frontend. In production for 2+ years, serving hundreds of internal users.
 - Built a real-time diagnostics application in C# with an extensible, configuration-driven architecture supporting per-client specifications.

PROJECTS

- Whiteboard: Real-Time Collaboration (Go, TypeScript, Protobuf, PostgreSQL)

Sep 2026 – now

 - Building an open-source whiteboard on a custom sync engine: server-sequenced per-property LWW merges with hybrid logical clocks, one actor goroutine per board, and edits group-committed to Postgres before ack.
 - Sent each client only its viewport via a spatial grid (~139 of 100k objects), halved p99 sync latency at 200 editors (254 → 124 ms), and cut full-history storage 5.5× to 31 B/edit.
- MilkRun: Real-Time Fleet Tracking (Java, Kafka, PostgreSQL/PostGIS, React)

Aug 2026

 - Architected a reactive telemetry pipeline with Spring WebFlux and reactor-kafka, streaming GPS events from 50 simulated delivery vans through Kafka to a React/Leaflet dashboard over Server-Sent Events.
 - Removed duplicate and out-of-order GPS events with an exact sliding-window deduplicator (8x a Bloom filter’s throughput) and per-van reorder buffers, committing Kafka offsets only after idempotent writes.
 - Predicted arrivals by map-matching GPS onto planned routes with PostGIS delay zones (~5 s mean error), and federated the live fleet with PostgreSQL history via Apache Calcite.
 - Shipped via GitHub Actions CD with automatic rollback; Prometheus and Grafana; 159 tests incl. Testcontainers.
- High-Performance Chat System (Ruby on Rails, Go, Redis, MySQL, Elasticsearch)

Nov 2025

 - Split a chat API into a Go write service (race-free sequence numbers via Redis atomic counters, async persistence through Sidekiq) and a Rails layer for reads and business logic, with Elasticsearch for full-text search.

EDUCATION

- B.Sc. in Computer Engineering · The American University in Cairo (AUC)

Aug 2017 – Jun 2022

 - Minors in Mathematics and Economics · Cum Laude (Honors).
 - Recipient of the Atef & Fofa Eltoukhy Family Endowed Scholarship (\$125K), awarded to 1 student every 5 years.
 - Represented Egypt at the 2017 Mediterranean Youth Mathematical Championship (bronze medal), qualifying through the Egyptian Mathematical Olympiad.
 - Graduation project: The Right Losses for the Right Gains (arXiv:2312.10854), a text-to-image architecture that outperformed state-of-the-art baselines, improving FID by 44%.

SKILLS

Languages	Java, Go, C#, Python, TypeScript, JavaScript, C++
Backend & APIs	Spring Boot, WebFlux, ASP.NET (MVC, Core), Node.js, Ruby on Rails, Apache Kafka, MQTT, Protobuf, REST
Cloud & DevOps	AWS (EKS, Lambda, RDS, DynamoDB, SQS), Docker, Kubernetes (Helm), Terraform, GitHub Actions, Jenkins
Data	PostgreSQL / PostGIS, MySQL, MongoDB, Redis (ElastiCache), Elasticsearch, Apache Calcite
Frontend	React, TypeScript, Next.js, Redux, Leaflet
Observability & Testing	Prometheus, Grafana, Micrometer, JUnit, Vitest, JMeter